AWS Application Load Balancer Project

Introduction

This project demonstrates how to configure an AWS Application Load Balancer (ALB) with multiple EC2 instances and target groups.

Different applications are hosted on separate EC2 instances, and the Application Load Balancer routes traffic based on URL paths such as `/mobile` and `/laptop`.

This is an extended architecture project where multiple EC2 servers are attached to each service for better scalability and high availability.

The Application Load Balancer distributes incoming traffic between multiple servers using the Round Robin algorithm.

This project helps in understanding:

- AWS Load Balancer
- Target Groups
- Path-Based Routing
- Round Robin Technique
- EC2 Integration
- High Availability Architecture
- Traffic Distribution

Architecture

```
                      User Browser
                           │
                           ▼
                Application Load Balancer
                           │
           ┌───────────────┼───────────────┐
           │               │               │
           ▼               ▼               ▼

       Home TG         Mobile TG       Laptop TG
           │               │               │
      ┌────┴────┐     ┌────┴────┐     ┌────┴────┐
      ▼         ▼     ▼         ▼     ▼         ▼

   Home-1    Home-2  Mobile-1  Mobile-2 Laptop-1 Laptop-2
    EC2       EC2     EC2       EC2     EC2      EC2
```

AWS Services Used

Service	Purpose
EC2	Hosting Applications
Application Load Balancer	Traffic Distribution
Target Groups	Route Requests
Security Groups	Allow HTTP Traffic

Project Structure

```
AWS Cloud
│
├── Application Load Balancer
│
├── Target Groups
│    │
│    ├── home-tg
│    │      ├── home-server-1
│    │      └── home-server-2
│    │
│    ├── mobile-tg
│    │      ├── mobile-server-1
│    │      └── mobile-server-2
│    │
│    └── laptop-tg
```

```
│         ├── laptop-server-1
│         └── laptop-server-2
│
└── Listener Rules
      ├── /mobile/*
      ├── /laptop/*
      └── default → home-tg
```

Step 1 — Launch EC2 Instances

Launch six EC2 instances:

Service	Instances
Home	2 EC2 Servers
Mobile	2 EC2 Servers
Laptop	2 EC2 Servers

Use:

- Amazon Linux 2023
- t3.micro instance type

Allow ports:

- SSH → 22
- HTTP → 80

Step 2 — Create Target Groups

Create three target groups:

Target Group	EC2 Instances
home-tg	home-server-1, home-server-2
mobile-tg	mobile-server-1, mobile-server-2
laptop-tg	laptop-server-1, laptop-server-2

Protocol:

- HTTP

Port:

- 80

Target Type:

- Instance

Step 3 — Create Application Load Balancer

Create an Application Load Balancer with:

- Internet Facing
- HTTP Listener on Port 80
- Attach all target groups

Step 4 — Configure Listener Rules

Add path-based routing rules:

Path	Target Group
/mobile/*	mobile-tg
/laptop/*	laptop-tg
Default	home-tg

Round Robin Technique

The Application Load Balancer uses the Round Robin algorithm to distribute incoming requests equally among multiple EC2 instances inside a target group.

Example:

- First request → home-server-1
- Second request → home-server-2
- Third request → home-server-1
- Fourth request → home-server-2

This technique helps in:

- Load Distribution
- Better Performance
- High Availability
- Reduced Server Load
- Improved Scalability

Application URLs

Home Page

```
http://application-lb-dns-name/
```

Mobile Page

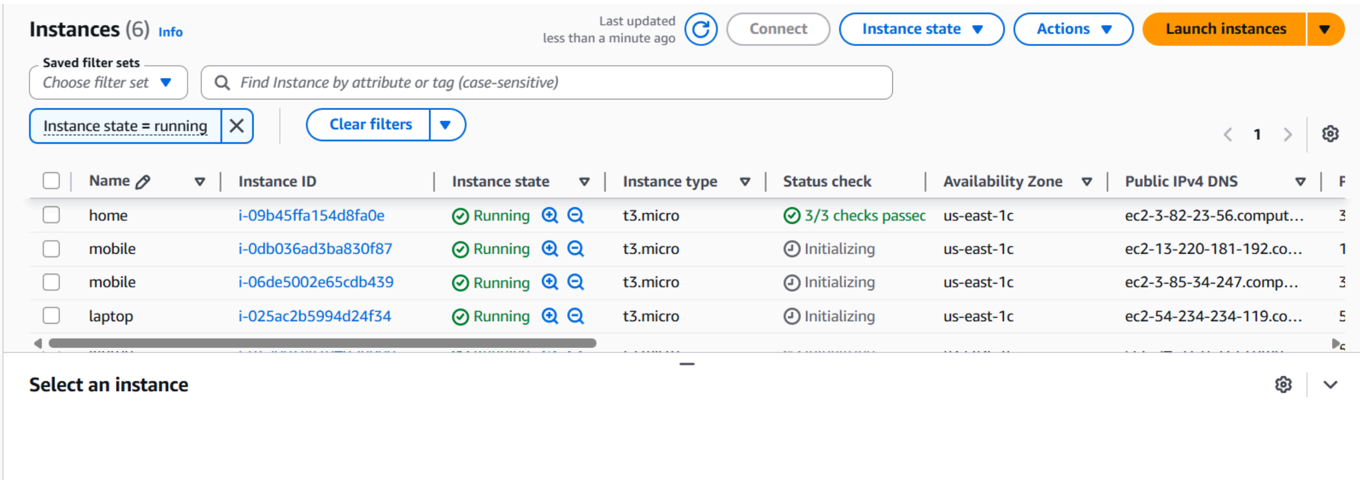
```
http://application-lb-dns-name/mobile/
```

Laptop Page

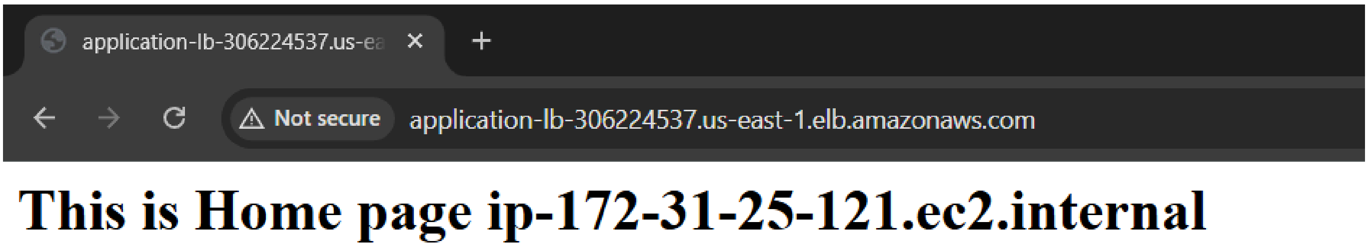
```
http://application-lb-dns-name/laptop/
```

Screenshots

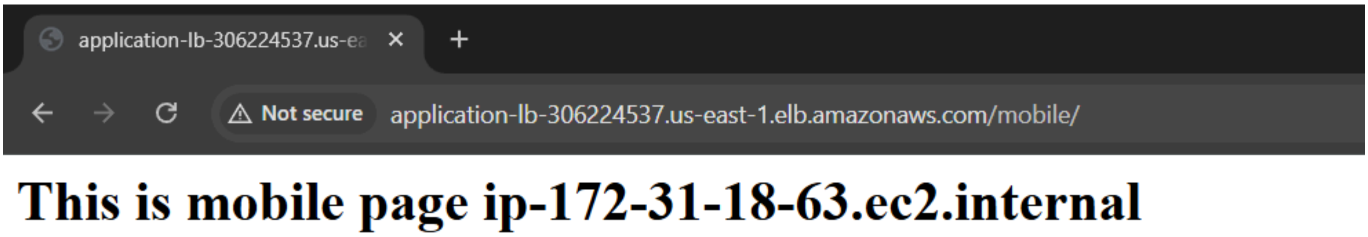
EC2 Running Instances



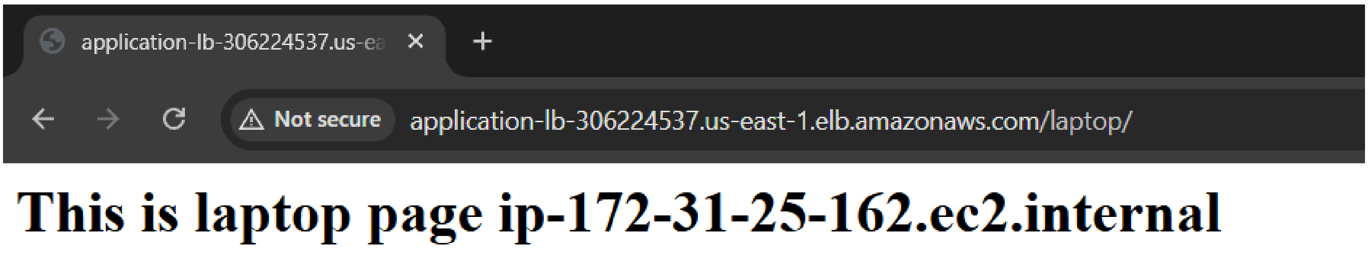

```
---
## Home Page Output
```



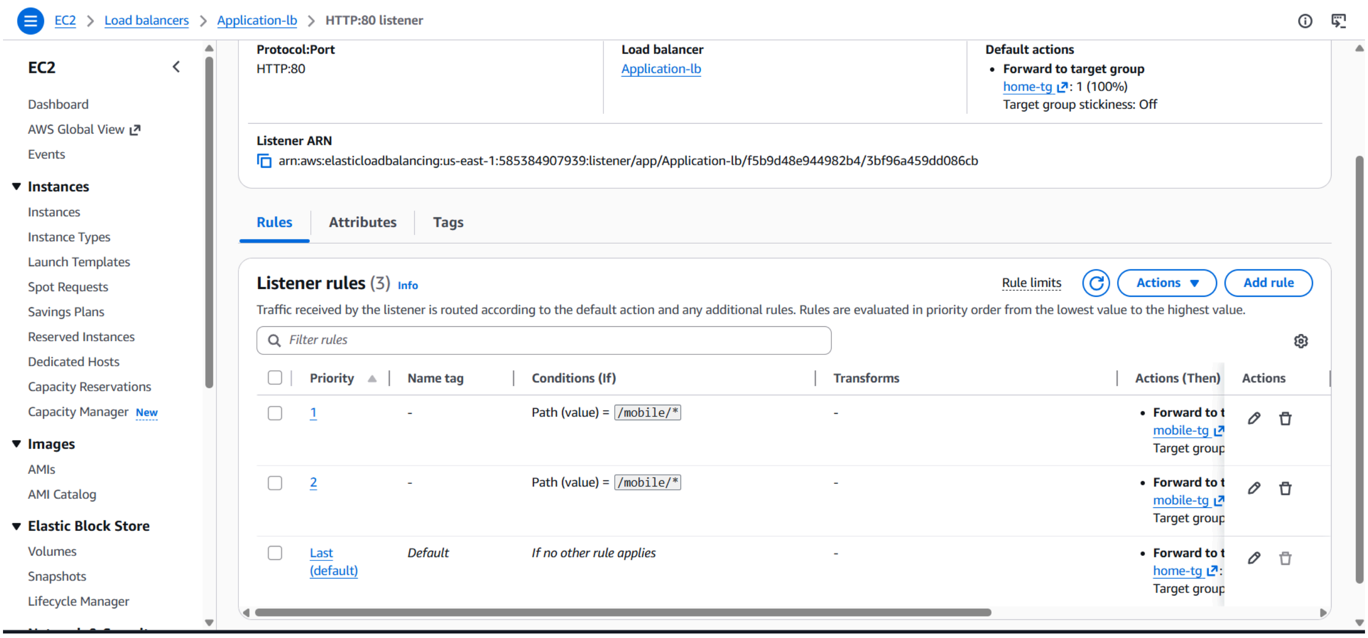

```
## Mobile Page Output
```



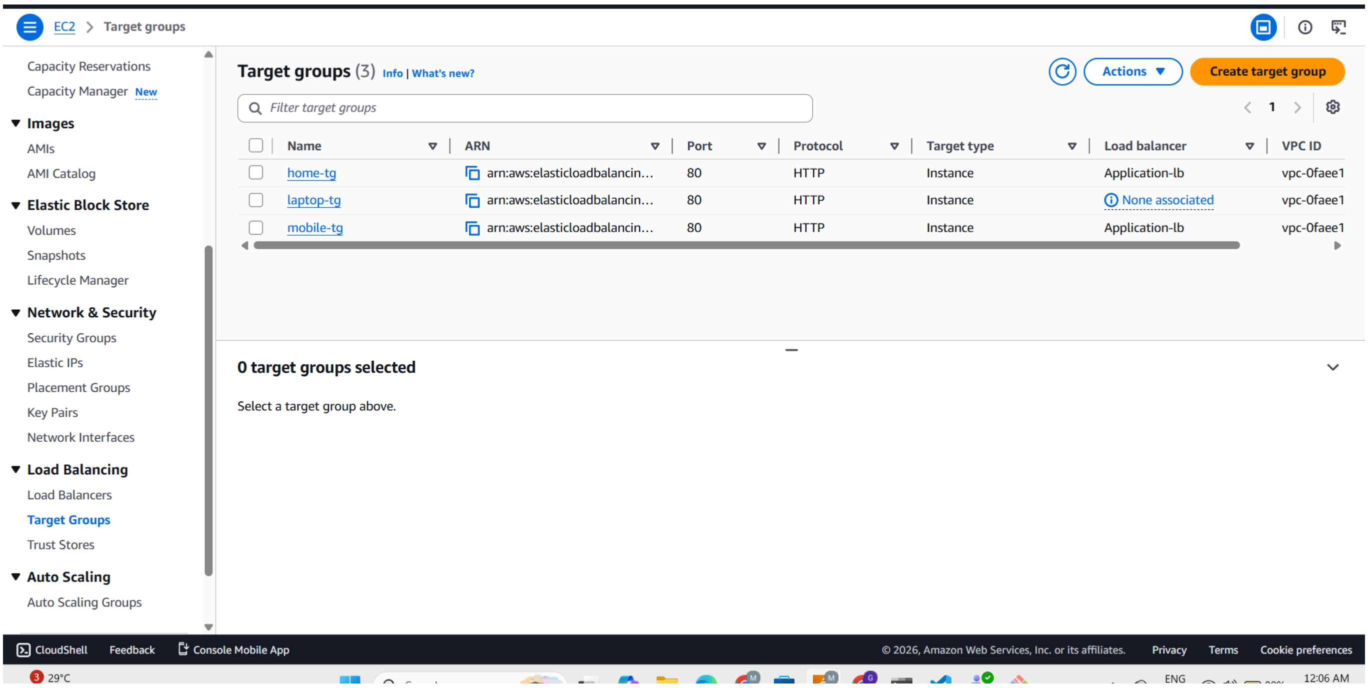

```
## Laptop Page Output
```




```
## Load Balancer Listener Rules
```




```
## Target Groups
```



Security Group Configuration

Type	Port
SSH	22
HTTP	80

Features

- Application Load Balancer
- Path-Based Routing
- Multiple EC2 Instances
- Round Robin Traffic Distribution
- High Availability
- Load Distribution
- Scalable Architecture
- Centralized Access

Future Improvements

- HTTPS SSL Configuration
- Auto Scaling Group
- Route53 Domain Setup
- Jenkins CI/CD Integration
- Docker Deployment

- Monitoring using CloudWatch

Author

Aniket Ajay Mulik

Fortune Cloud Technology AWS & DevOps Engineering

Summary

This project successfully demonstrates implementation of an AWS Application Load Balancer with multiple EC2 instances and target groups. Separate target groups were created for Home, Mobile, and Laptop services, with two EC2 servers attached to each group for high availability and scalability. The Application Load Balancer distributed traffic using path-based routing and the Round Robin technique, providing efficient traffic management and balanced server utilization in a cloud environment.
